

Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.



Dataset at a Glance

3,900

Purchases

18

Features

\$59.76

Avg. Spend

3.75

Avg. Rating

Key features: demographics, purchase details, shipping, discounts, review ratings, and subscription status.

Data Preparation in Python



01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Impute

Filled 37 missing Review Ratings with category median

03

Feature Engineering

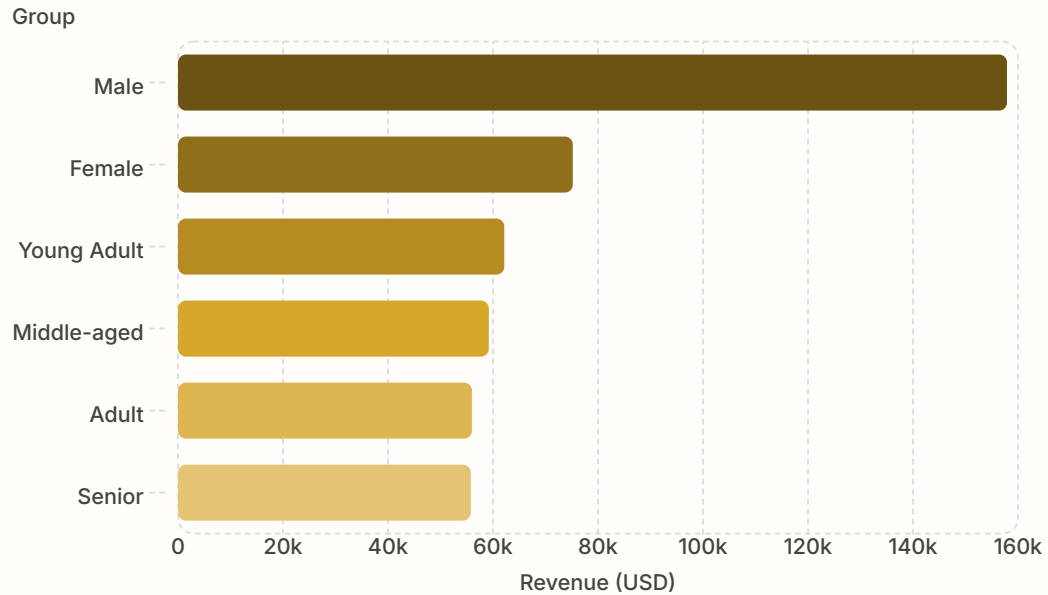
Created `age_group` and `purchase_frequency_days` columns

04

Load to PostgreSQL

Cleaned data exported to database for SQL analysis

Revenue Insights



Key Takeaways

- Male customers generate **2× more revenue** than female
- Young Adults are the **top revenue age group**
- Revenue is relatively balanced across adult age groups

Top Products by Category

Accessories

Jewelry (171) · Sunglasses (161) · Belt (161)

Clothing

Blouse (171) · Pants (171) · Shirt (169)

Footwear

Sandals (160) · Shoes (150) · Sneakers (145)

Outerwear

Jacket (163) · Coat (161)



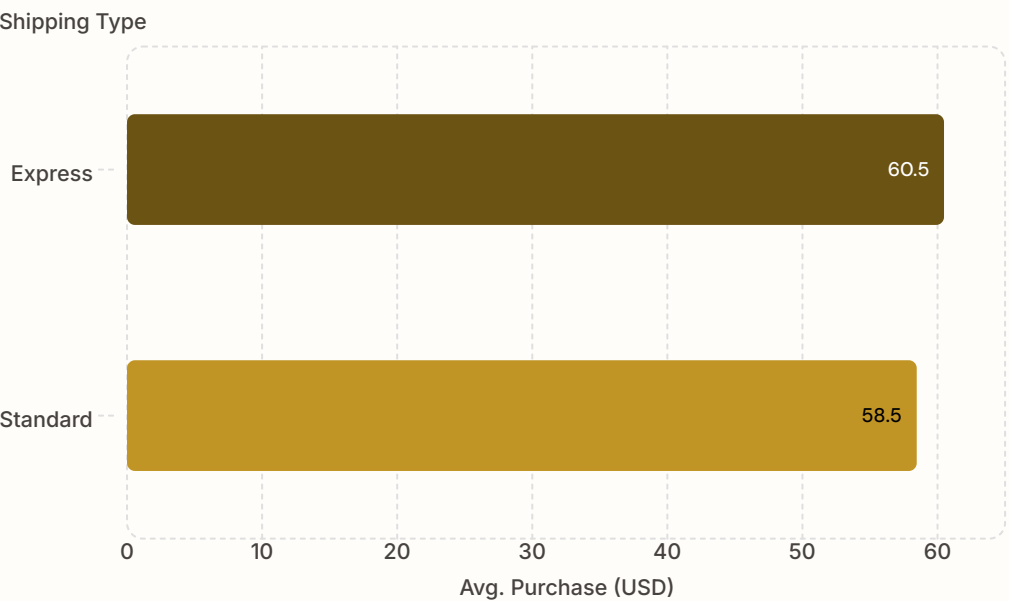
Discounts & Shipping

Discount-Heavy Products

Products with highest % of discounted purchases:

Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Shipping vs. Spend



Express shipping users spend **\$2 more** on average.

Customer Segmentation

Segment Breakdown



Loyal — 3,116

Highly engaged customers with more than five prior purchases; the core of repeat revenue and retention.

Returning — 701

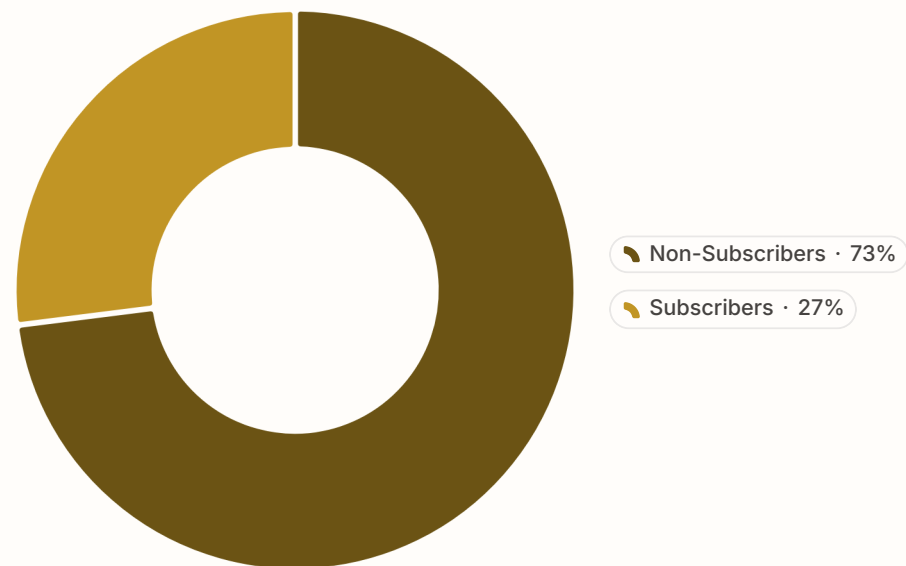
Customers with two to five purchases who show clear repeat behavior and strong potential for conversion to loyal status.

New — 83

First-time buyers who represent an important opportunity for onboarding, nurturing, and long-term growth.

i Even among discount users, 839 customers still spent **above average** — a high-value group worth prioritizing.

Subscription Behavior



Subscribers vs. Non-Subscribers

Metric	Subscribers	Non-Subscribers
Customers	1,053	2,847
Avg. Spend	\$59.49	\$59.87
Total Revenue	\$62,645	\$170,436

Repeat buyers (>5 purchases): **958 subscribed** vs. 2,518 not — loyalty drives subscription opportunity.

Power BI Dashboard



3.9K

Customers

\$59.76

Avg. Purchase

3.75

Avg. Rating

27% Yes

Subscribers



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow the Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users